

Lecture 2. Categorical Data Analysis

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Overview

- 1 Analysis of Two-Way Tables
- 2 Odds Ratios and Relative Risk
- 3 Introduction to Log-Linear Models

Two-Way Tables

Two-way tables are a very common way to view categorical data. We need a notation for them.

Rows= I

Columns= J

	Columns					
Rows	1	2	3	4	5	
1	n_{11}	n_{12}	n_{13}	n_{14}	n_{15}	n_{1+}
2	n_{21}	n_{22}	n_{23}	n_{24}	n_{25}	n_{2+}
3	n_{31}	n_{32}	n_{33}	n_{34}	n_{35}	n_{3+}
4	n_{41}	n_{42}	n_{43}	n_{44}	n_{45}	n_{4+}
	n_{+1}	n_{+2}	n_{+3}	n_{+4}	n_{+5}	n

Two-Way Tables

In this notation, each cell is n_{ij} where i is the *row* and j is the *column*.

The plus sign denotes marginal totals:

$$n_{i+} = \sum_{j=1}^J n_{ij}$$

$$n_{+j} = \sum_{i=1}^I n_{ij}$$

$$n = \sum_{i=1}^I n_{i+} = \sum_{j=1}^J n_{+j} = \sum_{i=1}^I \sum_{j=1}^J n_{ij}$$

Two-Way Tables

The data in the table can be collected in different ways:

- 1 Sample and classify into i and j , n fixed
- 2 Sample each j , classify into row i , e.g. n_{+1} is fixed
- 3 Each cell (i, j) in the table is an independent Poisson random variable

Two-Way Tables

Further, in two-way tables (especially in epidemiology), we set the table up such that the columns are the response variable and the rows are the independent variable.

Prospective Sampling

If the sampling is done within the rows/independent variable, then this is a prospective study. The response variable may not be observed.

Retrospective Sampling

If the sampling is done within the columns/dependent variable, then this is a retrospective study. Stratify on the response variable.

Two-Way Tables

Prospective Sampling Example

Exposure to dioxin may lead to cancer. Two comparable cities, but one has a dioxin-emitting plant. Sample in each city.

Retrospective Sampling Example

Draw samples of persons with cancer and a sample of persons without. Find out how much fresh fish they ate as children.

Example 2

Seat-Belt Use	Result of Crash	
	Fatality	Nonfatality
Yes	n_{11}	n_{12}
No	n_{21}	n_{22}

- Poisson: Catalog all accidents for a year and classify each on both dimensions, n is random
- Multinomial: Sample 100 police records and classify each, n is fixed
- Independent Multinomial: Sample 50 fatalities and 50 nonfatalities, n_{+1} , n_{+2} are fixed

NOTE: The last is an example of retrospective sampling. How would prospective sampling work in this setting?

2×2 Tables

Table			
Condition	Present	Absent	Row Total
Yes	n_{11}	n_{12}	n_{1+}
No	n_{21}	n_{22}	n_{2+}
Column Total	n_{+1}	n_{+2}	n

Two-Way Tables

Often the goal with data that can be presented in a two-way table is to see if there is an association between the two variables.

Consider the following 2×2 example.

	Disease		Row Total
	Yes	No	
Male	30	70	100
Female	20	80	100
Column Total	50	150	200

It seems as if there is an association between sex and disease.

Two-Way Tables

Another way to describe that association is to say that the percentages vary across the columns within the rows, or across the rows within the columns.

Here are the percentages in each column.

	Disease		Row Total
	Yes	No	
Male	0.3	0.7	
Female	0.2	0.8	
Column Total	0.25	0.75	

Two-Way Tables

Here are the percentages in each row.

	Disease		Row Total
	Yes	No	
Male	0.6	0.47	0.5
Female	0.4	0.53	0.5
Column Total			

Two-Way Tables

	Disease		Row Total
	Yes	No	
Male	30	70	100
Female	20	80	100
Column Total	50	150	200

What is n_{+1} ?

What is n_{2+} ?

What is $Pr(M, D)$?

What is $Pr(D|F)$?

Two-Way Tables

	Disease		Row Total
	Yes	No	
Male	30	70	100
Female	20	80	100
Column Total	50	150	200

What is n_{+1} ? 50

What is n_{2+} ? 100

What is $Pr(M, D)$? 0.15

What is $Pr(D|F)$? 0.20

Two-Way Tables

If the two variables are associated, then the marginal proportions won't be equal to the internal proportions. H_0 is that they are independent.

Recall our notation:

n_{ij} is the observed frequency in the cell in row i and column j .

n_{i+} is the row total

n_{+j} is the column total

If the two variables are unrelated, then any cell proportion is the product of the marginal proportions. Using the notation from last time:

$$\pi_{ij} = \pi_{i+}\pi_{+j}.$$

Two-Way Tables

Using our example, under independence, $Pr(D|M) = Pr(D)$.

We could also say, given only two categories (M & F),
 $Pr(D|M) = Pr(D|F)$.

Or, in words, $Pr(D)$ does not depend upon sex.

Two-Way Tables

This gives us a method for creating **Expected** counts if we want to test for independence.

Given these expected counts, we can use the Pearson X^2 and Likelihood Ratio Statistic G^2 to test the hypothesis.

Another way to write it, using our notation:

$$e_{ij} = np_{i+}p_{+j} = \frac{n_{i+}n_{+j}}{n}$$

Two-Way Tables

We can compare the statistic (X^2) estimated using these expected counts to a χ^2 distribution with $(I - 1)(J - 1)$ degrees of freedom.

Since the last value in any row or column is fixed. In the 2×2 case that means 1 df.

Degrees of Freedom

In the multinomial case, we have $k - 1$ degrees of freedom.

For example, when $k = 4$, once I know $\hat{p}_i$ for 3 of the categories, then the $\hat{p}_i$ for the fourth category is fixed (1 - sum of the other 3 categories).

Degrees of Freedom

In two-way tables, the degrees of freedom is a function of the null hypothesis.

For example, if the hypothesis fixes one margin, then once the the first three cells in each row are known, the fourth is also known.

1	2	3		n_{+1}
1	2	3		n_{+2}

Degrees of Freedom

If both margins are fixed, then once any cell is specified, the expected values are all set. Therefore, there is one degree of freedom.

16		20
		30
40	10	50

Another way to express the H_0 given by fixing both margins is to say that the H_0 is no association between the two factors.

Two-Way Tables

It is also possible to look at the individual deviations from a Pearson χ^2 .

$$\chi^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \frac{(O_{11} - E_{11})^2}{E_{11}} + \dots + \frac{(O_{IJ} - E_{IJ})^2}{E_{IJ}}$$

So, any cell has a deviation:

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Example 1

Let's look at an example. Counts of persons convicted of drunkenness in 2 London courts in first 6 months of 1970. SAS code:

```
proc freq order=data;  
  weight n;  
  tables gender*age / chisq nopercnt nocol  
    norow expected cellchi2;  
run;
```

Taken from Heiberger and Holland, p. 487.

Example 1

The FREQ Procedure

Table of gender by age

gender	age					
Frequency Expected Cell Chi-Square	0-29	30-39	40-49	50-59	>=60	Total
males	185 180.22 0.1269	207 209.78 0.0368	260 257.46 0.0252	180 178.31 0.016	71 77.237 0.5036	903
females	4 8.7814 2.6034	13 10.222 0.7551	10 12.545 0.5163	7 8.6885 0.3281	10 3.7635 10.335	44
Total	189	220	270	187	81	947

Statistics for Table of gender by age

Statistic	DF	Value	Prob
Chi-Square	4	15.2461	0.0042
Likelihood Ratio Chi-Square	4	12.6670	0.0130
Mantel-Haenszel Chi-Square	1	4.8961	0.0269
Phi Coefficient		0.1269	
Contingency Coefficient		0.1259	
Cramer's V		0.1269	

Sample Size = 947

Example 1

Now let's look at the example in R. The following code:

```
chisq.test(data.matrix(drunk))
```

Produces the following output:

```
Pearson's Chi-squared test
```

```
data: data.matrix(drunk)
```

```
X-squared = 15.2461, df = 4, p-value = 0.004217
```

```
Warning message:
```

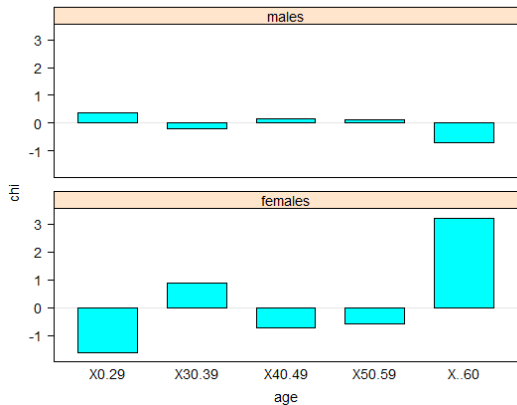
```
In chisq.test(data.matrix(drunk)) :
```

```
Chi-squared approximation may be incorrect
```

Example 1

We can also produce a graphic of the chi-square deviations:

chi deviations for drunk.dat



Two-Way Tables

This test relies upon the assumption that if the null hypothesis is true, then this statistic has a χ^2 distribution.

This can break down for tables with small cell sizes.

What is small? 5? 3?

Fisher's Exact Test

Pearson's X^2 relies on an approximation which can break down with small cell sizes.

An alternative is Fisher's Exact Test. It uses a hypergeometric distribution to determine the exact probability of the observed data occurring when the null hypothesis is true.

The hypergeometric distribution is a discrete distribution that describes the probability of k successes in n trials *without* replacement. $\frac{m}{N}$ is the probability of a success.

$$P(X = k) = \frac{\binom{m}{k} \binom{N-m}{n-k}}{\binom{N}{n}}$$

Fisher's Exact Test

Let's look at an example table.

	Disease		Row Total
	Yes	No	
Male	a	b	a+b
Female	c	d	c+d
Column Total	a+c	b+d	n

In this case, we want to calculate the probability of the observed data or more extreme values occurring under the null hypothesis.

Fisher's Exact Test

If we fix the margins at $a+c$ and $a+b$, then what is the probability that if we selected $a+c$ persons they would be distributed as a and c . We can use the hypergeometric distribution to calculate these probabilities.

$$p = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{a!b!c!d!n!}$$

We look at the probability of the observed data and any other possible tables with stronger associations. This can be either one- or two-tailed.

Fisher's Exact Test

Let's look at a simple 2×2 case (from Heiberger and Holland).

	Juvenile Delinquent	
	Yes	No
Glasses	1	5
No Glasses	8	2

What is the probability that the table would be this lopsided if wearing glasses was unrelated to delinquency?

Fisher's Exact Test

The probability of this configuration of the data under the null hypothesis [what is H_0 ?] is:

$$\frac{\binom{9}{1} \binom{7}{5}}{\binom{16}{6}} = 0.0236$$

There are a few tables that are less likely under the null hypothesis.

For example, 9 persons without glasses are delinquents and 0 without glasses are.

Fisher's Exact Test

These are all the possible tables. Why?

glasses	0		1		2		3		4		5		6	
	del	no.del	del	no.del	del	no.del	del	no.del	del	no.del	del	no.del	del	no.del
glasses	0	6	1	5	2	4	3	3	4	2	5	1	6	0
no.glasses	9	1	8	2	7	3	6	4	5	5	4	6	3	7

The tables have probability: 0.0009, 0.0236, 0.1573, 0.3671, 0.3304, 0.1101, 0.0105

Therefore, the total probability of the observed data or data even further from the null hypothesis if that hypothesis is true is $0.0236 + 0.0009 + 0.0105 = 0.0350$.

Therefore, we fail to accept the null hypothesis.

Example 2

Now let's look at the example in R. We have a *matrix* with our data:

```
> glasses
      delinq non.del
glasses      1      5
no.glasses   8      2
```

Example 2

We can request Fisher's exact test with the following code:

```
fisher.test(glasses)
```

Which produces the following output:

Fisher's Exact Test for Count Data

```
data:  glasses
```

```
p-value = 0.03497
```

```
alternative hypothesis: true odds ratio is not equal to
```

```
95 percent confidence interval:
```

```
 0.0009525702 0.9912282442
```

```
sample estimates:
```

```
odds ratio
```

```
0.06464255
```

Example 2

Now let's look at the example in SAS. We can request Fisher's Exact test with the following code:

```
proc freq data=glases order=data;  
tables glases*delinq /fisher ;  
weight count;  
run;
```

Example 2

Which produces the following output:

Fisher's Exact Test

```
-----  
Cell (1,1) Frequency (F)          1  
Left-sided Pr <= F                0.0245  
Right-sided Pr >= F                0.9991  
  
Table Probability (P)              0.0236  
Two-sided Pr <= P                  0.0350
```

Sample Size = 16

Example 2

Another approach is to adopt a Bayesian perspective. Beta-binomial:

```
# DATA
n1 <- 6 # glasses
y1 <- 1 # delinquents with glasses
n2 <- 10 # no glasses
y2 <- 8 # delinquents with no glasses

# SIMULATION
i <- 10000 # simulations
theta1 = rbeta(i, y1+1, (n1-y1)+1)
theta2 = rbeta(i, y2+1, (n2-y2)+1)
diff = theta1-theta2 # simulated diffs

# OUTPUT
print("Probability higher % delinquent with glasses than without:")
> print(mean(theta1>theta2))
[1] 0.0094
```

OR and RR: Setup

Row	Column	
	1	2
1	n_{11}	n_{12}
2	n_{21}	n_{22}

Setup

Row	Column	
	1	2
1	π_{11}	π_{12}
2	π_{21}	π_{22}

In these tables, the columns are the response variable, and the rows are the independent variable.

Setup

Now, define a conditional probability based on the latter table:

$$\pi_{j|i} = \frac{\pi_{ij}}{\pi_{i+}} = \frac{\pi_{ij}}{\pi_{i1} + \pi_{i2}}$$

In words, conditional on being in row i , the proportion in column j is $\pi_{j|i}$.

For example:

$$\pi_{1|2} = \frac{\pi_{21}}{\pi_{21} + \pi_{22}}$$

Setup: Example 3

	Disease	
	Yes	No
Male	0.15	0.35
Female	0.10	0.40

We want the conditional probability that you have disease given that you are male: $PR(D|M) = \pi_{1|1} = \frac{\pi_{11}}{\pi_{11} + \pi_{12}}$.

$$PR(D|M) = \pi_{1|1} = \frac{0.15}{0.15 + 0.35} = 0.3$$

Relative Risk

Now, define the relative risk.

$$\text{Relative Risk (Response Category 1)} = \frac{\pi_{1|1}}{\pi_{1|2}}$$

For example:

$$\frac{\pi_{1|1}}{\pi_{1|2}} = \frac{\Pr(D|M)}{\Pr(D|F)} = \frac{.3}{.2} = 1.5$$

Relative Risk

$$\text{Relative Risk (Response Category 2)} = \frac{\pi_{2|1}}{\pi_{2|2}}$$

For example:

$$\frac{\pi_{2|1}}{\pi_{2|2}} = \frac{\Pr(\bar{D}|M)}{\Pr(\bar{D}|F)} = \frac{.7}{.8} = 0.875$$

Relative Risk

An easy way to estimate the relative risk is:

$$\frac{n_{11}/n_{1+}}{n_{21}/n_{2+}}$$

The distribution of the relative risk is highly skewed. Therefore, it is better to estimate the variance on the log scale.

$$\hat{V} \left\{ \ln \left(\frac{n_{11}/n_{1+}}{n_{21}/n_{2+}} \right) \right\} = \frac{1 - \frac{n_{11}}{n_{1+}}}{n_{11}} + \frac{1 - \frac{n_{21}}{n_{2+}}}{n_{21}} = \frac{\Pr(\bar{D}|M)}{n_{11}} + \frac{\Pr(\bar{D}|F)}{n_{21}}$$

This is the variance of the logarithm of the Relative Risk(Response Category 1).

In-Class Problem

	Disease	
	Yes	No
Male	20	40
Female	30	30

What is the relative risk for men relative to women?

What is the variance of this estimate? Leave it on the log scale,

$$\hat{V} \left(\ln(\hat{RR}) \right).$$

In-Class Problem

	Disease	
	Yes	No
Male	20	40
Female	30	30

What is the relative risk for men relative to women? $\frac{20/60}{30/60} = \frac{.33}{.5} = 0.67$

What is the variance of this estimate? $\hat{V} \left(\ln(\hat{RR}) \right) = \frac{3}{60}$

Odds Ratio

Within row 1 the *odds* (not odds ratio) that the response is in column 1 instead of column 2 is defined as:

$$Odds_1 = \frac{\pi_{1|1}}{\pi_{2|1}} = \frac{\pi_{11}}{\pi_{12}}$$

From our example, this could be written:

$$\frac{\Pr(D|M)}{\Pr(\bar{D}|M)} = \frac{\Pr(D|M)}{1 - \Pr(D|M)}$$

Continuing the example, the odds that a man will have the disease are $\frac{.3}{1-.3} = .43$. For women, this odds are $\frac{.2}{1-.2} = .25$

Odds Ratio

Those are the odds. The ratio of the odds is called the *odds ratio*.

$$\theta = \frac{\pi_{1|1} / \pi_{2|1}}{\pi_{1|2} / \pi_{2|2}}$$

Remember that $\pi_{1|1} = \frac{\pi_{11}}{\pi_{11} + \pi_{12}}$. So we can also write it in the following manner:

$$\theta = \frac{\frac{\pi_{11}}{\pi_{11} + \pi_{12}} / \frac{\pi_{12}}{\pi_{11} + \pi_{12}}}{\frac{\pi_{21}}{\pi_{21} + \pi_{22}} / \frac{\pi_{22}}{\pi_{21} + \pi_{22}}} = \frac{\pi_{11} / \pi_{12}}{\pi_{21} / \pi_{22}} = \frac{\pi_{11} \pi_{22}}{\pi_{21} \pi_{12}}$$

Hence the name *cross-product ratio*.

Odds Ratio Misinterpretation Example

Take care with interpretation. Here is a headline from a newspaper in the UK:

MailOnline



[Home](#) | [News](#) | [U.S.](#) | [Sport](#) | [TV&Showbiz](#) | [Femail](#) | **[Health](#)** | [Science](#) | [Money](#) | [RightMind](#)

[Health Home](#) | [Health Directory](#) | [Health Boards](#) | [Diets](#) | [MyDish Recipe Finder](#)

Statins can weaken muscles and joints: Cholesterol drug raises risk of problems by up to 20 per cent

- **Statins are the most widely prescribed drugs in Britain**
- **However, many complain of muscle pain and joint weakness**

By JENNY HOPE

PUBLISHED: 22:42, 3 June 2013 | **UPDATED:** 00:00, 4 June 2013

Odds Ratio Misinterpretation Example

David Spiegelhalter:

"Unfortunately, the '20%' is factually incorrect - the study on which this story is based claims that taking statins increased the risk of muscle problems from 85% to 87%."

Odds without statins: $\frac{0.85}{0.15} = 5.67$

Odds with statins: $\frac{0.87}{0.13} = 6.69$

Odds ratio: $\frac{6.69}{5.67} = 1.18$

Odds ratio misinterpreted as increase in probability.

Odds Ratio

Characteristics of the Odds Ratio

- 1 $\theta = 1$ is independence
- 2 $\theta > 1$ subjects in row 1 are more likely to have the response
- 3 $\theta < 1$ subjects in row 2 are more likely to have the response
- 4 Values farther away from 1 indicate a stronger relationship
- 5 $\ln(\theta) = 0$ is independence since $\ln(1) = 0$
- 6 $\ln(\theta)$ is symmetric around 0

Odds Ratio

We estimate the odds ratio using:

$$\hat{\theta} = \frac{n_{11}n_{22}}{n_{21}n_{12}}$$

The variance of $\ln(\hat{\theta})$ can be estimated as:

$$\hat{V} \left\{ \ln(\hat{\theta}) \right\} = \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}$$

Example 4

Physicians Health Study Data				
Myocardial Infarction				
	Heart Attack	No Attack	Total	P_{i+}
Placebo	189	10845	11034	0.017
Aspirin	104	10933	11037	0.009

The relative risk of heart attack is $\frac{189/11034}{104/11037} = \frac{0.017}{0.009} = 1.82$.

The odds ratio is:

$$\hat{\theta} = \frac{(189)(10933)}{(104)(10845)} = 1.83$$

Example 4

The variance is determined for $\ln(\hat{\theta}) = 0.60543771$.

$$\hat{V} \left\{ \ln(\hat{\theta}) \right\} = \frac{1}{189} + \frac{1}{104} + \frac{1}{10845} + \frac{1}{10933} = 0.01509006$$

Example 4

Form the confidence interval with care. Do all the work on the logarithmic scale!

$$1 \quad \sqrt{V\{\ln(\hat{\theta})\}} = 0.12284.$$

$$2 \quad 1.96 * 0.12284 = .2408.$$

$$3 \quad 0.6054 - 0.2408 = .3646 \text{ and } 0.6054 + 0.2408 = 0.8462. \text{ This is the 95\% confidence interval on the logarithmic scale.}$$

$$4 \quad \text{Exponentiate to get back to the scale of } \hat{\theta}. \text{ Therefore, } (e^{0.365}, e^{0.846}) = (1.440, 2.331).$$

Example 4

We can do this work in R. Here I'm using the *epiR* package:

```
library(epiR)
```

```
myocard<-matrix(data=c(189,104,10845,10933),nrow=2)
```

```
myocard<-as.table(myocard)
```

```
epi.2by2(myocard)
```

Example 4

Which produces the following output:

	Disease +	Disease -	Total	Inc risk *	Odds
Exposed +	189	10845	11034	1.713	0.01743
Exposed -	104	10933	11037	0.942	0.00951
Total	293	21778	22071	1.328	0.01345

Point estimates and 95 % CIs:

Inc risk ratio	1.82 (1.43, 2.31)
Odds ratio	1.83 (1.44, 2.33)
Attrib risk *	0.77 (0.47, 1.07)
Attrib risk in population *	0.39 (0.15, 0.62)
Attrib fraction in exposed (%)	44.99 (30.22, 56.63)
Attrib fraction in population (%)	29.02 (17.25, 39.12)

* Cases per 100 population units

Example 4

We can do the same thing in SAS using PROC FREQ:

```
proc freq data=myocard order=data;  
tables aspirin*myocard / cmh;  
weight count;  
run;
```

Example 4

Which produces the following output:

Estimates of the Common Relative Risk (Row1/Row2)

Type of Study	Method	Value	95% Confidence Limits	
Case-Control (Odds Ratio)	Mantel-Haenszel	1.8321	1.4400	2.3308
	Logit	1.8321	1.4400	2.3308
Cohort (Col1 Risk)	Mantel-Haenszel	1.8178	1.4330	2.3059
	Logit	1.8178	1.4330	2.3059
Cohort (Col2 Risk)	Mantel-Haenszel	0.9922	0.9892	0.9953
	Logit	0.9922	0.9892	0.9953

Total Sample Size = 22071

Relationship between Relative Risk and Odds Ratio

Remember the table from earlier:

	Disease		Row Total
	Yes	No	
Male	a	b	a+b
Female	c	d	c+d
Column Total	a+c	b+d	n

Relationship between Relative Risk and Odds Ratio

We can write the relative risk as:

$$RR = \frac{a/(a+b)}{c/(c+d)}$$

If a is small, then $a/(a+b)$ is approximately a/b . The same applies to the denominator.

$$RR = \frac{a/(a+b)}{c/(c+d)} \approx \frac{a/b}{c/d} = \frac{ad}{bc} = OR$$

But only if the condition is rare!

Ordinal Variables

We have been ignoring any possible ordering of the categories.

The methods here assume nominal categorical variables.

Some variables contain information about order: ordinal variables.

The tests we have described may be helpful in determining independence for ordinal variables.

Still, ignoring the ordering in the data is a loss of information. Decrease of efficiency. There are tests that use knowledge of the ordering.

Log-Linear

Log-linear models are a useful tool for analyzing data in contingency tables.

Possible to explore relationships as was done with Pearsons X^2 . But more flexible!

Today, we will just look at how these models relate to other methods we have seen for discrete data.

Log-Linear

Recall that for a Two-Way Table, we wrote the probability mass function in this way:

$$P(\mathbf{n}; \pi) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J \pi_{ij}^{n_{ij}}$$

Log-Linear

We could rewrite this such that the cell probabilities are modeled as counts:

$$\begin{aligned} P(\mathbf{n}; \pi) &= \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J \left(\frac{m_{ij}}{n} \right)^{n_{ij}} = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J m_{ij}^{n_{ij}} n^{-n_{ij}} \\ &= \frac{n! n^{-n}}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J m_{ij}^{n_{ij}} \end{aligned}$$

Log-Linear

Under the independence hypothesis, we can write the *expected* counts as:

$$m_{ij} = n\pi_{i+}\pi_{+j} = n \left(\frac{m_{i+}}{n} \right) \left(\frac{m_{+j}}{n} \right) = n^{-1} m_{i+} m_{+j}$$

This says that the expected cell counts are proportional to the product of the marginal counts. With a little algebra, we can take logarithms of the RHS:

$$\ln(m_{ij}) = -\ln(n) + \ln(m_{i+}) + \ln(m_{+j})$$

Log-Linear

The latter equation looks like a regression model with main effects and no interaction terms.

Again, under *independence*, that model is correct. An interaction term implies lack of independence. Please explain why?

Typically, these models are written in this form:

$$\ln(m_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y$$

Log-Linear

They are subject to the constraint that they are deviations around a grand mean which must sum to 0 (effect coding – -1 1):

$$\ln \left(n \prod_{i=1}^I \pi_{i+} \right) = \sum_{i=1}^I \lambda_i^X = 0$$

There are $1 + (I - 1) + (J - 1) = I + J - 1$ free parameters.

Log-Linear

That is the model under the assumption of independence. We can add interactions. This model will have perfect prediction (Expected=Observed in every cell) and is called the *saturated* model.

$$\ln(m_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$$

There is an additional restriction imposed:

$$\sum_{i=1}^I \lambda_{ij}^{XY} = \sum_{j=1}^J \lambda_{ij}^{XY} = 0$$

The saturated model has 1 free parameter for λ , $I - 1$ free parameters for row main effects, $J - 1$ free parameters for column main effects, and $(I - 1)(J - 1)$ free parameters for interactions. For example, 1 interaction term in a 2×2 table.

$$I + J - 1 + (I - 1)(J - 1) = IJ.$$

Log-Linear

The general approach for log-linear model is to fit the saturated model and then eliminate terms to see if the model still fits.

For example, eliminating the XY interaction term in the model above implies independence.

More on this in the future.